National College of Ireland

BSHCIFSC2, HCCOMP2, BSHCIFSC2_AE_SD, BSHCIFSC2_AE_NT

Release Date: February 2, 2026
Due Date: March 27, 2026

Lecturers:
Charlyn Rosales
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Software Engineering (H6SWE)

Continuous Assessment (CA): Assignment 50%

Weight: 50%
The assignment will be marked out of 50.

Instructions: This is a group project to be completed in groups of 3 to 4 with a mix of group and individual deliverables. Students will develop a computer-based system linked to the student's chosen topic that demonstrates student's ability to:

- Identify and model system requirements using UML.
- Perform system behaviour modelling through appropriate diagrams.
- Apply Scrum methodology to plan and manage development.
- Develop a functional prototype of core system features.

SUBMISSION DETAILS:
Submission to Moodle for each deliverable as per project description below

TURNITIN: All report submissions will be electronically screened for evidence of academic misconduct (i.e., plagiarism and collusion)

Late submissions will not be penalized if the student applied for an extension through NCI360 and it was approved.

Students may be asked to attend a viva examination to assess their understanding of the work submitted.

Instructions:

- Project deliverables should be completed in groups of 3-4 people.
- Team disruptions should be managed within the team, if agreement cannot be reached, please contact the lecturer as soon as possible.
- The project is composed of several deliverables to be completed over several submissions prior to the final submission date.
- The project work should be carried out as homework and in specified labs.
- CASE Tools: Visual Paradigm (downloadable) or Draw IO must be used to complete the relevant deliverables.
- Several of the milestones should be presented during the lab times (as indicated on Moodle) and later submitted to Moodle.
- The final submission will include a video presentation where the students will present their prototype through the use of documentation (these will be listed in detail below).
- For each submission there should be one submission per team – stating clearly all the team members involved. There may be individual components which should be labelled by the respective student who constructed this component.

Deliverables:

Checkpoint 1 (Requirements and Analysis Modelling using UML)
Due: 11.59pm Friday, February 20th

1. **Use Case Diagram *(Team Submission)***

- Use case diagram/model (1 for the entire team, the diagram must use at least one extends and includes relationships each). Use case diagram should include the following:
    - o At least 4 use cases
    - o At least 1 includes relationship
    - o At least 1 extends relationship
    - o A generalization relationship between actors

2. **Detailed Use Case (UC) Descriptions *(Individual)***
    - Each team member must submit one fully detailed use case
    - Using the template for use cases from the model.
    - Please see Moodle for an example of a use case template.
    - Use Case Modelling Principles
        - Relevant Actors are Identified
        - Relevant UC's are identified
        - UC's provide value to the actor

- UC descriptions may based on the template provided or any form that details the interaction between the actor & the use case.
- The UC should be described in detail
- The activation is correctly written
- Main flow is a sequence of logical transactions
- The main flow is enumerated
- Alternate flows are described and labelled correctly

3. **Conceptual Class Diagram of a Chosen Use Case (Team Submission)**

   The conceptual class diagram should demonstrate the use of all of the following. This component is a single submission for the entire team:
   - Classes with Attributes
   - Associations
   - Multiplicity
   - Relationships
   - Navigability & Association class (optional)

(20 marks)

Checkpoint 2 (System Behaviour Modelling using Sequence Diagrams)
Due: 11.59pm Sunday, March 8th

1. **System Sequence Diagrams and Contracts (Team Submission)**
   The team should create **two System Sequence diagrams** for selected use case (other types of sequence diagrams will receive a zero) to describe the flow within use cases (two members of the team will create this diagram, should be named next to the diagram on the Moodle submission).

2. **System Operation Contract (Team Submission)** The remaining team member/s who didn't create the diagram should develop a contract for the system operations in the sequence diagram (again, label these/this student/s). The contract must include:
   - Operation Name
   - Cross-reference to Use Case
   - Preconditions
   - Postconditions

(10 marks)

Checkpoint 3 (Agile Development & Prototype)
Due: 11.59pm Sunday, March 27th

Agile Requirements

Each team should submit the following:

1. **Product Backlog**
   Incorporating all the use cases prioritized in Product Backlog Items (PBIs)

2. **Sprint Backlog**
   For one 2-week sprint:
   - 2-3 user stories only each must include:
     - Derived from the original 'Use Cases'.
     - Summarized as user stories with detailed acceptance criteria and estimates (this is a requirement to receive full marks).
     - For a team of 4 people, two people will be responsible for a user story each, and the other two will be responsible for the respective acceptance criteria of each user story.

3. **Sprint Tasks**
   Each student must:
   - Breakdown one user story into implementation tasks.
   - Submit their own set of tasks for a separate PBI.
   - This is a mark given for the entire team.

4. **Definition of Done**
   Team must clearly state what makes a backlog item "Done" (ie. Code complete, tested merged to main branch, through a series of unit, integration, and system tests etc.)

Mock-up (Simple Prototype) Requirements

- Implement 2-3 core use cases only, you don't need to develop the full system.
- It can be console-based, desktop, or web-based.
- Front-end only is fine. You can use drag-and-drop tools like Figma, InVision Studio, Adobe XD, Canva, or any similar tool for simplicity.
- If you prefer, you can also use development environments such as NetBeans, web app frameworks, or any other IDE. The goal is to choose the tool that helps you finish on time.
  Note: The goal is to show your idea and how it works, not to make a full working system. Focus on clarity, usability, and completing it on time.
- GitHub repository required. Must use Main branch & Development branch
- GitHub may have at least 2 milestones reflecting project progress

- README.md may include system overview and setup instructions (Configuration Manual).

5. **Final presentation** (PPT presentation + MS Teams Meeting Video Recording)
    - Duration: not more than minutes per group.
    - Presentation must include:
        - System overview
        - Demonstration of Prototype
        - Explanation of how scrum was applied
        - One challenge encountered and how it was addressed
    - Each slide must clearly indicate student contributor names. Students who do not do this will not receive equal marks from their presentation.
    - Video and GitHub repository link submission instructions:

      The URL of the **video** and **GitHub** repository link must be provided in the presentation and make sure that examiners have access. The examiners reserve the right to conduct live mini presentations with a selected sample of students, during which students may be required to answer questions pertaining to their tasks. Live mini presentations may also be scheduled if additional clarification is deemed necessary or if concerns arise regarding the submission.

      Please record your video presentation using **Microsoft Teams**, ensure each student's **face** and **student ID card** are clearly visible at the start of the recording. Failure to submit the video and failure to follow the instructions will result in Checkpoint 3 deliverables not being assessed.

(20 marks)

Grading Rubric below:

Marking Scheme			
Topic	**Grade Item**	**Marks**	**Criteria**
Use Case Diagram 8%	Use Case Diagram	2	Correct System Boundary and Actors
	Use cases	2	At least 4 meaningful use cases
	Relationships between use cases in UML	2	Correct use of <<include>> or <<extend>>
	Logical relationships and clarity	2	Correct and Logical modelling of relationships and clarity across the Use Case diagram.
Use Case Descriptions 6%	Main Flow	2	Clear main flow, logical and numbered steps
	Pre/Post Conditions & Alternate Flows(s)	2	Valid pre-conditions & post conditions, appropriate alternate flow(s)
	Overall Structure	2	Clarity, structure, and completeness
Conceptual Class Diagram 6%	Classes & Attributes	2	Appropriate classes & attributes
	Relationships & Multiplicity	2	Correct relationships and multiplicity
	Associations	2	Proper use of associations.
2 System Sequence Diagram & Contracts 10%	Format	2	Correct System Sequence Diagram format
	System Operations	2	Correct and relevant system operations identified
	Alignment with Use Case	2	Logical sequence aligned with the use case.
	Operation Contract	4	Correct operation signature, clear pre-conditions, post conditions (state changes described).
Agile & Prototype 20%	Product Backlog	8	All use cases represented as PBIs, logical prioritization, and clear descriptions
	Sprint Backlog		2-3 well-formed user stories, clear, testable acceptance criteria, and realistic estimation.
	Sprint Tasks		Logical breakdown of work, clear ownership, and appropriate task granularity.
	Prototype Implementation	7	2-3 use cases implemented, functionality matches acceptance criteria, GitHub usage (branches, milestones, and README)
	Final Presentation	5	Report was professionally presented using PPT and MS Teams recording. including clear system explanation, proper demonstration of the prototype, explanation of scrum application, and reflection on challenge encountered.